

Indian Institute of Engineering Science and Technology, Shibpur
B.Tech. (CST) 3rd Semester Mid-Term Examination, October 2022
Discrete Structures (CS - 2101)

Time: 2 hours

Full Marks: 30

Answer all questions.

Write all parts of the same question together.

1.
 - (a) Name the principle used for proving the uncountability of the set of real numbers.
 - (b) What is the name of the proof that proves $p \rightarrow q$ is true based on the fact that p is false?
 - (c) The logic representations of universally and existentially quantified statements use the following operators, respectively: (i) Conjunction and bi-implication (ii) Implication and disjunction (iii) Disjunction and conjunction (iv) Implication and conjunction
 - (d) Given the predicates $G(x)$: x is a gold ornament; $S(x)$: x is a silver ornament; $P(x)$: x is precious, what will be the logical representation of the statement. "Gold and silver ornaments are precious."
 - (e) Show which of the following propositions is a tautology. (i) $p \wedge q \rightarrow p$ (ii) $p \vee (q \rightarrow p)$ (iii) $p \wedge (p \rightarrow q)$ (iv) $p \rightarrow (p \rightarrow q)$
 - (f) Name the rule of inference used in this argument: "If I go for a balanced diet, then I will be fit. If I will be fit, then I will remain healthy. Therefore, if I go for a balanced diet, then I will remain healthy."
 - (g) A function is defined by mapping $f : A \rightarrow B$ such that A contains m elements and B contains n elements and $m \leq n$. Find the number of one-to-one functions.
 - (h) Are these system specifications consistent? "The system is in multiuser state if and only if it is operating normally. If the system is operating normally, the kernel is functioning. The kernel is not functioning or the system is in interrupt mode. If the system is not in multiuser state, then it is in interrupt mode. The system is not in interrupt mode."

[1 + 1 + 1 + 2 + 2 + 1 + 1 + 3 = 12]
2. Use quantifiers and predicates with more than one variable to express these statements.
 - (a) There is a student in this class who can speak Spanish.
 - (b) Every student in this class plays some sport.
 - (c) Some student in this class has visited Alaska but has not visited Hawaii.
 - (d) All students in this class have learned at least one programming language.
 - (e) Every student in this class has chatted with at least one other student in at least one chat group.
 - (f) There is a student in this class who has taken every course offered by one of the departments in this school.

[1 × 6 = 6]
3. Use rules of inference to show that the hypotheses "Andy works hard," "If Andy works hard, then he is a dull boy," and "If Andy is a dull boy, then he will not get the job" imply the conclusion "Andy will not get the job."

[5]
4.
 - (a) Use a proof by contradiction to show that there is no rational number r for which $r^3 + r + 1 = 0$ [Consider the standard form of the rational number].
 - (b) Use proof by contraposition to show that if n is an integer and $n^3 + 5$ is odd, then n is even.

[4 + 3 = 7]